Mechanical Systems and Signal Processing 235 (2025) 112951

Contents lists available at ScienceDirect

Mechanical Systems and Signal Processing

journal homepage: www.elsevier.com/locate/ymssp

Full Length Article

Fault detection and safety risk evaluation of lithium-ion batteries based on confidence interval of Gaussian process regression for real-world application

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ARTICLE INFO

Communicated by Simon Laflamme

Keywords:
Electric vehicle
Lithium-ion batteries
Fault detection
Early warning
Safety evaluation

ABSTRACT

Electric vehicles (EVs) have emerged as a significant contributor to reducing carbon emissions and promoting sustainable development. Accurately evaluating the safety risk of lithium-ion batteries is the key to ensuring the safe and reliable operation of EVs. However, the occurrence and progression of battery faults are uncertain, especially in complex practical scenarios. To this end, we design a novel fault detection and safety evaluation framework featuring adaptive thresholds based on confidence interval (CI) of Gaussian process regression (GPR). This framework enables battery faults to be detected, located, and tracked in real-world scenarios. The effectiveness of the proposed method is verified on a dataset of 1046 cells from nine faulty EVs, two battery modules and twenty-eight battery systems. The average precision and false positive rate of fault detection results are 0.99 and 0.07, respectively. Compared with the battery management system alarm, the proposed method provides 22.3–240.1 h and 2–28 s early warnings for developmental and sudden faults, respectively. Furthermore, our method accurately tracks the evolution trend of faults and enables the visualization of battery safety risks. This work highlights the potential and generalizability of GPR-based CI as adaptive thresholds for monitoring and diagnosing battery faults.

1. Introduction

The development of transportation electrification technology has emerged as a promising solution to address the challenges posed by environmental pollution, reducing fossil fuel usage, and achieving carbon neutrality [1]. Electric vehicles (EVs), which are equipped with lithium-ion batteries, have gained significant attention due to their ability to reduce traffic emissions [2]. However, the safety of lithium-ion batteries has become one of the main factors hindering the development of EV technology. According to the National New Energy Vehicle Big Data Alliance, over 50 % of EV fires originate from battery faults [3]. For example, China recorded an

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https://doi.org/10.1016/j.ymssp.2025.112951
Received 19 December 2024; Received in revised form 12 May 2025; Accepted 2 June 2025
Available online 5 June 2025
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annual average of 31 fire incidents, with 36.9 % and 26.2 % of incidents caused by sudden ignition and charging of batteries, respectively [4]. Moreover, in comparison to fires in internal combustion engine vehicles, battery-related fires are less predictable and pose greater risks [5]. Therefore, accurate battery safety risk monitoring and early fault detection prior to irreparable catastrophic failures such as thermal runaway are crucial for enabling timely maintenance of EVs and preventing life and property damage caused by fires.

Battery faults can arise from various causes, including material defects during the manufacturing process, exposure to extreme operating conditions, and improper use [6]. While a battery pack can tolerate minor faults in individual cells and still comply with regulations, prolonged usage can exacerbate these faults, leading to the development of severe faults (SFs) [7]. Under extreme conditions, such as high temperatures or excessive mechanical stress, unpredictable thermal runaway can occur [8]. To identify potential thermal runaway safety hazards, researchers have explored many early fault detection methods, which can be broadly divided into three categories: rule-based, model-based, and data-driven methods [9]. Compared with temperature, voltage, which is more sensitive to battery faults, is usually used to detect faults [10]. Rule-based methods determine whether a battery is faulty by analyzing whether the collected battery voltage or temperature data exceeds a preset threshold. However, defining an appropriate threshold based solely on prior knowledge becomes challenging in complex practical scenarios. Setting the threshold too low can result in excessive sensitivity, increasing the likelihood of erroneous fault detection. On the other hand, setting the threshold too high may delay meaningful fault detection, potentially compromising safety and reliability. Model-based methods accurately simulate the internal state of batteries and compare the measured values and key battery information obtained by models to determine the occurrence of faults [11]. However, despite demonstrating effectiveness in laboratory scenarios, model-based approaches pose challenges when applied to real-world applications characterized by dynamic operating conditions, diverse vehicle types, and limited data quality [12].

In recent years, data-driven approaches leveraging statistical analysis and machine learning techniques have gained popularity in the field of battery diagnostics [13]. These methods rely solely on battery operating data and exhibit strong nonlinear fitting capabilities without in-depth knowledge of battery electrochemistry [14]. For instance, Gaussian process regression (GPR) has attracted much attention in the field of battery diagnosis due to its good nonlinear fitting capabilities [15]. However, thus far, no research has delved into the potential of utilizing GPR estimated confidence intervals (CIs) to address the battery fault detection, and only estimating the mean value has been focused. In addition, with advancements in network communication and cloud storage technology, the utilization of field data from EVs has become feasible. For example, Zhang et al. [16] proposed a dynamic autoencoder model that leverages charging snippets of EVs for fault detection. Li et al. [17] constructed a hybrid neural network to estimate the internal resistance of batteries and evaluate battery safety in real-world EVs by analyzing abnormal internal resistance occurrences. Nevertheless, current studies still face three major limitations. First, some algorithms initially have strict data condition requirements, and they rely on stable conditions to extract features. However, the accessibility of specific data for EVs can be limited due to privacy concerns and storage constraints. Second, there is room to improve the real-time performance and lead time of fault warning. Some methods only focus on the charging process for battery fault detection, the possibility of faults occurring during driving is overlooked, resulting in untimely or even meaningless detection. Third, many existing methods utilize a fixed threshold for fault detection, which is determined by analyzing the diagnostic results of malfunctioning EVs via these methods. However, these methods cannot adaptively adjust the threshold based on the specific operating conditions of different EVs. This limitation poses safety risks, as fixed thresholds may prove ineffective in monitoring vehicles with unknown operating conditions.

To fill the above research gaps, we propose a novel probability-based safety evaluation framework for real-time battery fault detection and safety risk assessment. In this framework, we leverage the CI derived from GPR as an adaptive fault threshold. Although GPR is popular in the field of battery health assessment due to its excellent nonlinear fitting performance, no studies have explored the value of its CI as an adaptive fault threshold. As a probability distribution, GPR-based CI is naturally suitable for evaluating the uncertainty law of fault occurrence and evolution [18]. Moreover, a strategy is proposed to train the model once and estimate CI independently for each cell, which effectively solves the problem of false positives (FPs) caused by the traditional CI-based threshold. In addition, to meet the goal of quantifying battery safety, such as monitoring the state of charge and state of health, the proposed framework incorporates a dynamic safety risk evaluation equation to continuously track changes in a battery's safety risk. This work highlights the promising potential and practicality of GPR-based CI as an adaptive threshold for addressing uncertain problems, especially in the context of battery safety evaluation.

The remainder of this paper is organized as follows. Section 2 describes the field data used for validating. Section 3 introduces the methodologies of fault detection and safety risk assessment. Section 4 evaluates and discusses the validation results. Section 5

Table 1
Dataset information of real-word EVs.

EV label	Cell number	Temperature probe number	Sampling frequency (s)	Elapsed Time (days)	Sampling points	Fault type
EV #1	156	33	30	207	41,851	Developmental
EV #2	89	20	30	112	13,389	Developmental
EV #3	8	8	10	75	74,132	Developmental
EV #4	96	32	10	44	22,338	Developmental
EV #5	89	32	10	5	27,201	Sudden
EV #6	88	32	30	131	37,153	Sudden
EV #7	96	32	10	24	56,835	Sudden
EV #8	92	32	10	7	32,714	Sudden
EV #9	96	32	10	100	96,531	Sudden

concludes the paper.

2. Description of battery data for real-world EVs

Real-time monitoring of a battery based on the data recorded by a BMS plays a crucial role in ensuring safe EV operation. To support the advancement and validation of the approach, we have collected the processed operational data of nine faulty EVs collected from the China Southwest Branch of the National Monitoring and Management Center for New Energy Vehicles and Enterprises. The operational data contain essential information pertaining to battery performance, such as time, voltage, current, state of charge (SOC), cell voltage and probe temperature. Detailed data are recorded in Table 1 (these EVs are labeled EV #1-EV #9). The types of battery faults experienced by these EVs can be broadly divided into two categories: developmental (EV #1-EV #4) and sudden (EV #5-EV #9). Except for EV #1, other EVs were eventually involved in thermal runaway accidents, as confirmed by engineers. It is worth mentioning that EV #3 uses lithium iron phosphate batteries, and the other EVs are equipped with ternary batteries.

To illustrate the unpredictable nature of battery faults, Fig. 1 presents plots showing the voltage variations in the battery pack for each EV, alongside the SF alarm information provided by the BMS. The typical phenomena of internal short circuit (ISC) faults, voltage drops are reflected in developmental fault EVs [19]. Following the sudden failure of an EV, the affected battery experiences a significant voltage drop in a very short time. Under such circumstances, the recorded sensor data becomes invalid due to the extreme conditions and damage caused by the incident. Moreover, existing built-in BMS algorithms are not timely or sensitive for detecting faults. Notably, the voltage anomalies of the two faulty cells in EV #1 unexpectedly returned to normal after they appeared for a period. Possible explanations are that the protective material in the battery continues to support the battery structure, avoiding further contact between the negative and positive electrodes, or that the melting of the lithium dendrites interrupts the ISC process [20,21]. This phenomenon indicates that the safety risk of the battery has rebounded. However, if the safety risk cannot be accurately determined, it is still prudent to consider the battery to be at a low safety level. Proceeding with maintenance and recycling efforts without a precise assessment of the safety risk may lead to avoidable economic losses. In addition, due to the protection of data privacy by the manufacturer, the sampling frequency of the uploaded data of EV#1, #2 and #6 is limited to 30 s. Only the discharge data are available for EV #1 and #2. These limitations render approaches that rely on high-fidelity data and charging segments inapplicable. To address these challenges, we propose a probability-based safety assessment framework designed for diverse EV operation data in real-

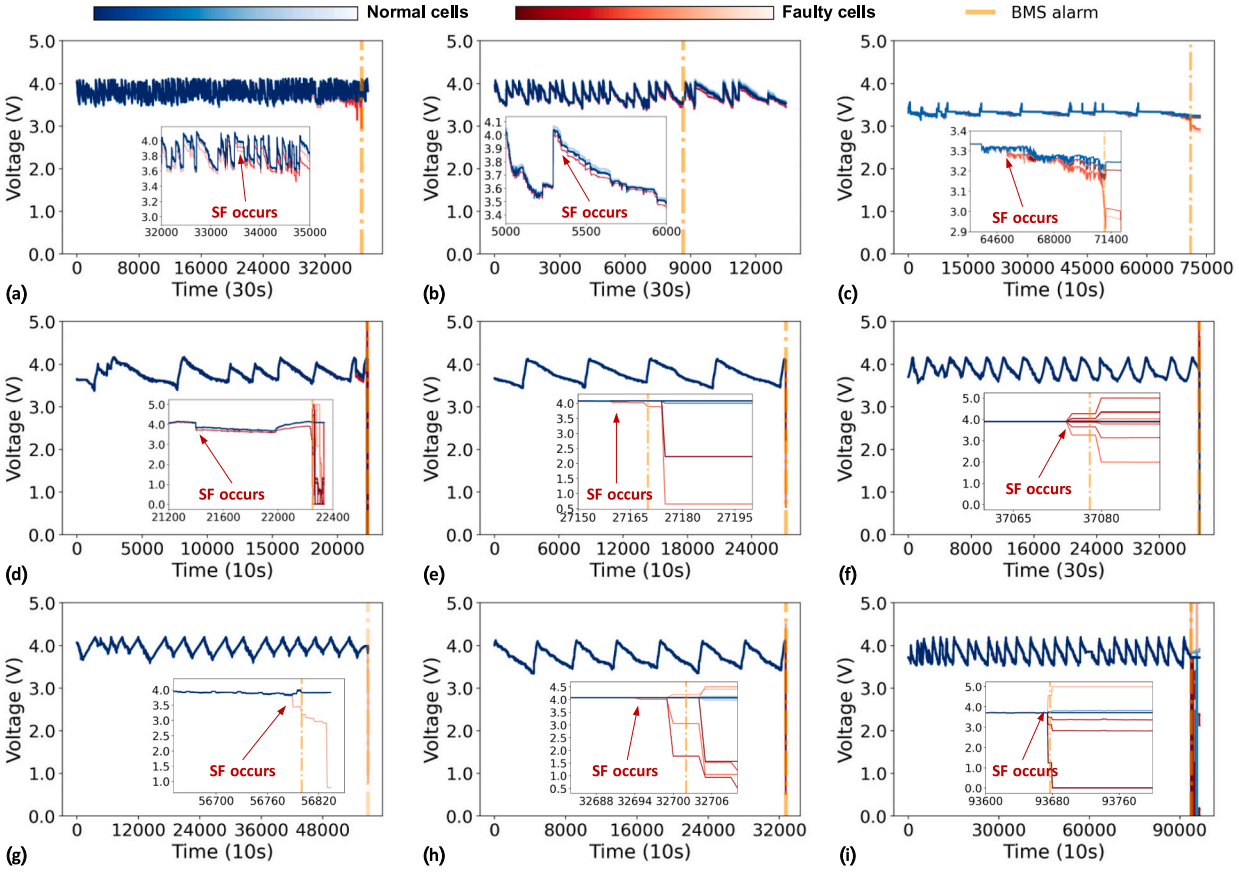


Fig. 1. Voltage variation in EV operation data. (a)–(i) are EV #1- EV #9. The enlarged figures in each Fig. show the moment when the SF occurs.

world scenarios.

3. Methodology

We design a safety evaluation framework for real-time diagnosis in field scenarios, as shown in Fig. 2. The framework aims to achieve early detection and accurate localization of faulty cells, as well as to continuously quantify the cell safety. Moreover, the framework adopts the strategy of cloud-based molding and regular updating (Fig. 2a). This strategy uses advanced communication technology to avoid the waste of computing power on the vehicle. Two main components of the framework are probability-based fault detection and safety risk assessment with a dynamic threshold (Fig. 2b), which are introduced in detail below.

3.1. Gaussian process regression

To realize safety evaluation based on probability, a powerful machine learning-based technique, GPR, is applied in this study. GPR is a nonparametric regression technique based on the GP that enables uncertainty estimation in posterior predictions and exhibits notable advantages in non-linear scenarios [22]. The GP is a stochastic process that is completely specified by its mean function $m(x)$ and covariance function (also referred to as its kernel function)$k(x, x')$ and is denoted as:

$$f(x) = GP(m(x), k(x, x')) \tag{1}$$

The mean and covariance functions are fundamental components of the GP model and can be represented as follows:

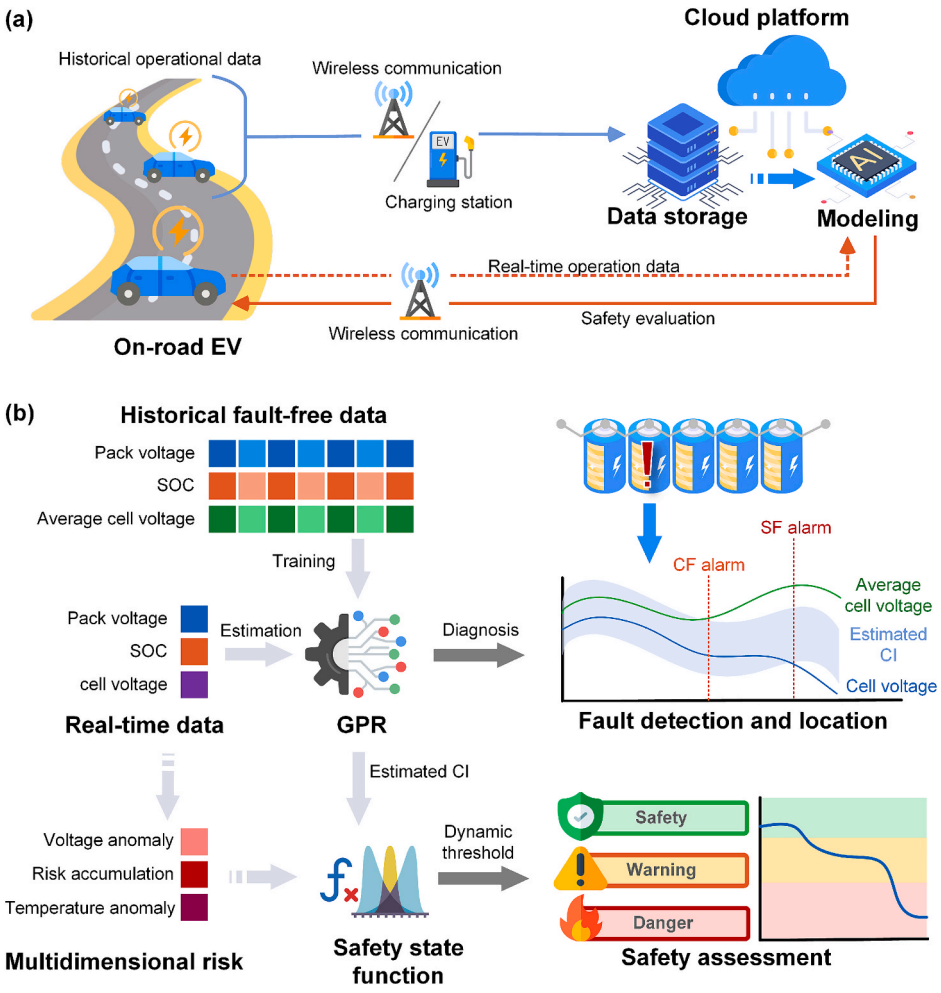


Fig. 2. Overview of the proposed safety evaluation framework. (a) Data acquisition and model building. Historical operation data are transmitted to the cloud platform through wireless communication or charging device collection. On-road EVs interact with the cloud model through wireless communication to obtain real-time safety evaluation results. (b) Diagram of the evaluation procedure, including input choice, fault detection and location, and safety assessment.

$$m(x) = E(f(x)) \tag{2}$$

$$k(x, x') = E[(m(x) - f(x))(m(x') - f(x'))] \tag{3}$$

Commonly, the zero-mean function ($m(x) = 0$) and the squared exponential (SE) kernel function (also known as the radial basis function) are employed in GPR. The SE kernel function can be defined as:

$$k_{SE}(x, x') = \exp(-\frac{1}{2}(x - x')^T \Theta^{-2}(x - x')) \tag{4}$$

where Θ is a length scale parameter. Due to the complex operation environment of an EV, it is necessary to consider the influence of noise on the output $y(x)$ of GPR, which can be expressed as:

$$y(x) = f(x) + \varepsilon \tag{5}$$

where x represents a feature of feature set X. ε represents the noise $N(0, \sigma_n^2)$, which includes data noise and measurement errors during data acquisition. Therefore, for any given training set $D = (X, y)$, the prior distribution derived from GPR can be written as:

$$y \sim \mathcal{N}(0, K(X, X) + \sigma_n^2 I_n) \tag{6}$$

where $K(X, X)$ is the covariance matrix, and the element that makes up this matrix is $K_{ij} = k(x_i, x_j)$. I_n is a unit matrix with n dimensions.

To further improve the accuracy of the estimation, the hyperparameters θ in the covariance function k need to be solved by minimizing the negative log marginal likelihood (*NLML*). The *NLML* can be expressed as $NLML = -\log p(y|x, \theta)$. To infer the optimal hyperparameters in the *NLML*, the adaptive moment estimation (Adam) algorithm is selected as the optimizer in this paper. Compared with those of other optimization algorithms, Adam has a faster convergence speed and stronger learning effect [23].

For estimation or prediction tasks, it is often assumed that the test and training sets follow the same Gaussian distribution. When the test data x^* are entered into GPR, the estimated result y^* is obtained. The joint prior distribution of y and y^* can be described as:

$$\begin{bmatrix} y \\ y^* \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K(X, X) + \sigma_n^2 I_n & K(X, x^*) \\ K(x^*, X) & K(x^*, x^*) \end{bmatrix} \right) \tag{7}$$

Then, according to Bayes' theorem, the test output y^* can be calculated from the following posterior distribution:

$$p(y^* | x^*, X, y) = N(y_m^*, y_v^*) \tag{8}$$

The output y^* contains two parts, the mean y_m^* and the variance y_v^*, which represent the best estimation and the uncertainty of the estimation of GPR, respectively. Thus, the CI can be calculated by $[y_m^* - z\sqrt{y_v^*}, y_m^* + z\sqrt{y_v^*}]$. z represents the confidence level of the CI.

3.2. Safety risk function

Evaluating the real-time safety risk of a battery is indispensable in a complete fault diagnosis system. According to the relationship between safety and risk, the safety risk function of a battery can be defined as the inverse proportion of the risk state of the battery, that is,

$$f_{safety}(x) = \frac{1}{f_{risk}(x)} \tag{9}$$

where f_{safety} and f_{risk} represent the safety risk of the battery and the risk state of the battery, respectively. x represents the states used to describe the battery behavior in different dimensions, for example, voltage, temperature, current, and internal resistance.

To set a reasonable value range to evaluate the safety risk of the battery, the value range is set to 0 to 1 in this paper by referring to the setting of the state of health (SOH). When the fault risk of the battery becomes infinitely large, that is, when the battery experiences thermal runaway, the safety of the battery should be 0, indicating that it is absolutely unsafe. When no fault occurs and the fault risk of the battery is 0, its safety should be 1, indicating that the battery is absolutely safe. Therefore, the safety risk function can be rewritten as

$$f_{safety}(x) = \frac{1}{g(x) + 1} \tag{10}$$

where $g(x)$ is a function of the battery fault risk, which is defined as $\geqslant 0$. The following relationship is needed:

$$f_{0\%} = 0 = \lim_{g \to \infty} \frac{1}{g + 1} \text{ indicates that thermal runaway occurs} \tag{11}$$

$$f_{100\%} = 1 = \lim_{g \to 0} \frac{1}{g + 1} \text{ indicates that no fault occurs} \tag{12}$$

To determine whether a common fault will evolve into a serious fault, it is necessary to define the value of SFs occurring in the numerical range of the safety risk function. This approach is inspired by the numerical setting of the end of life (EOL) of batteries in EV applications. When the SOH drops to 0.8, the battery is considered to have reached its EOL and needs to be replaced and recycled [24]. Therefore, when $f_{safety} = 0.8$, a SF is considered to occur.

For practical safety/risk assessment, a bell curve is often used as the safety/risk function. Common bell curves include Gaussian distributions, Cauchy distributions and logarithmic Gaussian distributions [25]. For the convenience of calculation, the safe risk equation in Equation (11) is rewritten in a general form similar to that of the Cauchy distribution, that is,

$$f_{safety}(x) = \frac{1}{\alpha(h(x) - \beta)^2 + 1} \tag{13}$$

where $h(x)$ represents the fault risk of a battery, and its value is not limited to $\geqslant 0$. α and β are constants used to control safety risk changes. Combined with the previous setting of numerical conditions, α and β can be solved by the following relation:

$$1 = \frac{1}{\alpha(h(x_1) - \beta)^2 + 1} \tag{14}$$

$$0.8 = \frac{1}{\alpha(h(x_{0.8}) - \beta)^2 + 1} \tag{15}$$

from which the parameters are solved

$$\beta = h(x_1) \tag{16}$$

$$\alpha = \frac{0.25}{(h(x_{0.8}) - h(x_1))^2} \tag{17}$$

where $h(x_1)$ and $h(x_{0.8})$ represent the boundaries for no fault and SF, respectively. In this study, the boundaries for the voltage-based evaluation are set to the CI obtained by real-time estimation (see Section 4.2). Moreover, the boundaries for fault-free and severe faults for the temperature-based evaluation are set to 55 °C and 90 °C, respectively [26]. Finally, the safety risk function can be expressed as

$$f_{safety}(x) = \frac{1}{0.25\left[\frac{h(x) - h(x_1)}{h(x_{0.8}) - h(x_1)}\right]^2 + 1} \tag{18}$$

When evaluating the safety risk of the battery from n dimensions, the final *Score* will be obtained by calculating all the values of f_{safety}, which can be expressed as follows:

$$Score = a_1 f^1_{safety} + a_2 f^2_{safety} + \ldots + a_n f^n_{safety} \tag{19}$$

where $a_1 \sim a_n$ represent the weights of the impact of different factors on battery safety ($a_1 + a_2 + \ldots + a_n = 1$). When the score is closer to 1, the battery is safer.

3.3. Safety evaluation framework

Our framework is specifically designed to address the challenge of real-time safety monitoring without the need for additional sensors. By following the principle of constructing a model that outputs the normal voltage based on the BMS recorded data, the proposed framework allows for the identification of battery faults by comparing the measured voltage against this model. Fig. 3

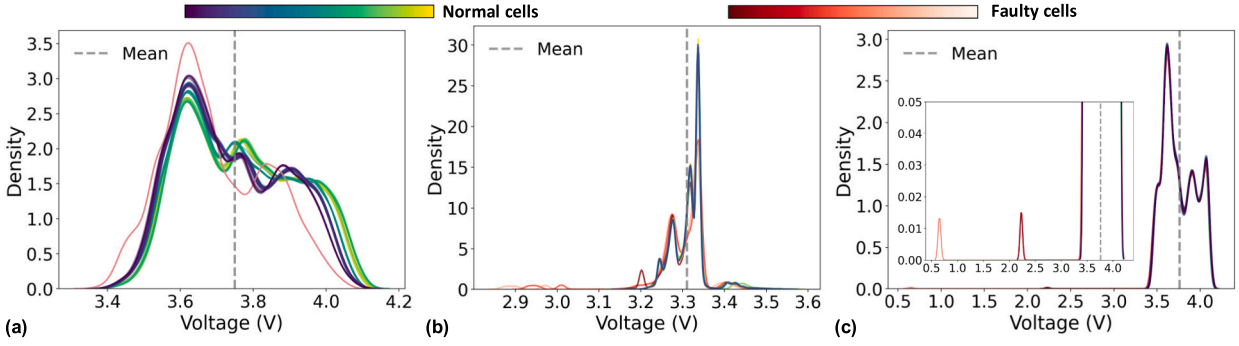


Fig. 3. Density analysis of cell voltages. (a) EV#2 (ternary batteries with developmental fault). (b) EV#3 (lithium iron phosphate batteries with developmental fault). (c) EV#5 (ternary batteries with sudden fault).

illustrates the results of the density analysis of the cell voltage. It can be seen that when the battery is normal, the voltage distribution is similar despite the inconsistency. When a fault occurs, the voltage will obviously deviate from the normal voltage distribution. Although some existing studies also exploit the deviation of fault voltage from normal distribution to detect faults, their performance in practical applications is limited (details are discussed in Section 4.1). In addition, to avoid inefficiency caused by modeling multiple cells independently, we adopt a model training scheme to diagnose all cells simultaneously. To build the fault-free model, we select three parameters that exhibit a high correlation with cell voltage: battery pack voltage, average cell voltage, and SOC (see Supplementary Note 1 for voltage indicator selection).

During fault monitoring, the trained model outputs the corresponding CI for each cell voltage, which serves as the fault detection threshold. Once the measured cell voltage exceeds the CI, the cell is considered to have an SF. The SF alarm information is sent to the user and monitoring center so that timely emergency measures can be implemented. To further exploit the CI as an adaptive threshold, we introduce the real-time average cell voltage for determination. If the average cell voltage exceeds the CI of a cell, the inconsistency of the cell is too large, which is regarded as a common fault (CF) occurs. The CF alarm information is regarded as an accumulation of safety risks because excessive voltage inconsistency is often an indicator of imminent accelerated aging or even ISC faults [27]. This information is transmitted to the manufacturer or regulator for safety monitoring.

To enrich the simple binary fault detection results of true or false faults, we propose a safety assessment method with dynamic thresholds. This method allows the user or regulator to track the progression of faults over time, gain a deeper understanding of the safety risk of batteries, and repair the battery pack in a timely manner to avoid the occurrence of thermal runaway accidents. Based on the CI estimated for each cell, an independent evaluation of each cell in the battery pack is achieved. Moreover, the safety evaluation is conducted from three different perspectives to ensure comprehensive evaluation outcomes. First, the voltage anomaly is diagnosed based on the measured cell voltage and estimated CI. Second, the cumulative risk of failure is evaluated. The risk is defined as the voltage difference between the average cell voltage exceeding the boundary of the CI (the upper and lower boundaries for overvoltage and undervoltage, respectively), and the voltage difference accumulated over time is treated as the cumulative risk. Finally, the temperature anomalies are evaluated by analyzing the maximum probe temperature. This approach is chosen due to the limited knowledge available regarding the specific arrangement of battery packs in the research EVs. By selecting the maximum probe temperature, we aim to capture the most critical safety conditions among all the tested cells. This conservative approach allows us to obtain safety assessment results that prioritize the highest level of safety.

4. Results and discussion

To evaluate the performance of the proposed method, the data of 822 cells from faulty EVs and battery modules are used for validation. All presented validation results are averaged over 50 repeated experiments to avoid the impact of chance events on the performance evaluation. All experiments are run on a personal computer (Intel i7-14700 K CPU, 32 GB RAM and Nvidia RTX 4060Ti). The mean absolute error is calculated for the GPR model evaluation, which is mathematically expressed as

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{20}$$

where y_i and $\hat{y}_i$ are the measured voltage and estimated voltage, respectively. n denotes the number of sample points. Moreover, precision and false-positive (FP) rate are selected as metrics to evaluate the performance of fault detection, as follows

$$\text{Precision} = \frac{TP}{TP + FP} \tag{21}$$

$$\text{FP rate} = \frac{FP}{FP + TN} \tag{22}$$

where TP, TN and FP represent corrective positive-predicted data, corrective negative-predicted data, and wrong positive-predicted

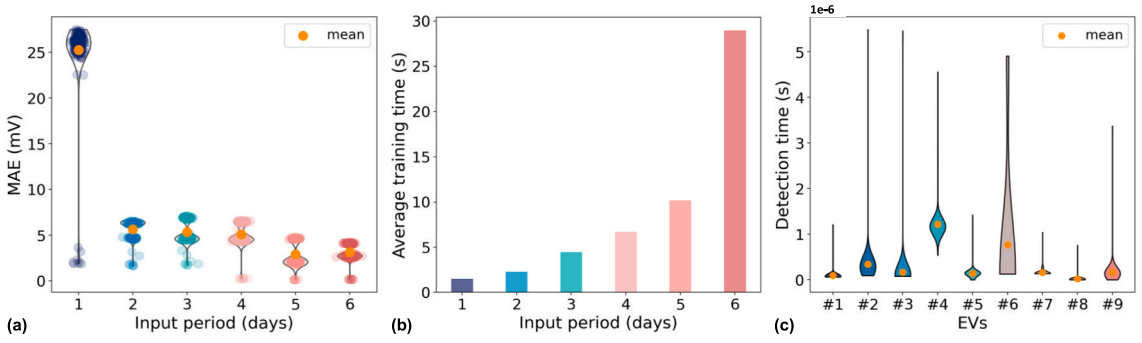


Fig. 4. Voltage estimation performance with different input periods. (a) The estimation errors. (b) Average training time. (c) Fault detection time.

data, respectively. In addition, the early warning time compared to the BMS alarm is used as a metric of the timeliness of fault detection.

4.1. Probability-based fault detection and location

To achieve accurate and effective fault detection, it is necessary to obtain a reliable fault threshold. However, it is difficult to effectively evaluate the performance of a GPR model with output thresholds based only on CIs. To this end, we evaluate the performance of the GPR model based on the accuracy of voltage estimation. Moreover, the influence of various user habits on the model performance is compared (the amount of data in the same period is inconsistent), and the results are shown in Fig. 4. It can be seen that when the input period is 5 days or more, the improvement of model accuracy is no longer obvious, but the increase of training time is still significant. Therefore, to balance model accuracy and computational cost, data with a period of 5 days is selected as model input in this study (average values of MAE and training time are 2.89 mV and 10.19 s, respectively). In addition, the fault detection times achieved by the proposed method for each EV are depicted in Fig. 4(c). Notably, the detection time recorded by the proposed methodology for a single row of data falls below 1.5e-6 s. This outcome underscores the feasibility of a real-time fault detection application and highlights the negligible time delay introduced by the detection process.

After evaluating the model performance and training model, another important factor to analyze is the CI. To determine the CI that is most suitable for the battery system fault threshold in the field scenario, we evaluate the fault detection performance of five CIs with different confidence levels (80 %, 85 %, 90 %, 95 % and 99 %). To illustrate the underlying principle of our proposed fault detection and localization method, we take EV #1 as an example and show the detection results in Fig. 5. This provides a visual representation of how our method effectively identifies and pinpoints faults within a battery system.

Fig. 6 shows the detection results for each faulty EV with different CI (3-σ denotes the CI based on the traditional 3-σ rule [28]). It can be observed that each CI achieves satisfactory results for EVs with sudden faults (EV #5-#9). This is because the voltage values of faulty cells deviate significantly from the normal range. For EVs with developmental faults (EV #1-#4), the difference in detection performance of different CIs is obvious. The detection results of 3-σ CI are not acceptable in these EVs with lower precision and higher FP rate. The reason is that 3-σ CI is the method that only relies on voltage inconsistency to detect faults. It is easy to misjudge the voltage with large inconsistency but fault-free as a fault. In contrast, the proposed adaptive threshold is calculated independently for each cell, which effectively solves the problem of FPs caused by inconsistency fluctuations (A visual comparison is shown in Supplementary Fig. S3). Comparing the five proposed CI-based thresholds, when the confidence level is 95 % or greater, it is too large for early fault detection in real-world scenarios. The CI of 95 % and 99 % ignore the soft ISC that occurs in EV #2 (the precision and FP rate are both 0). When the confidence level is 80 %, it leads to negative performance (particularly, reducing the precision of EV #3). Therefore, to detect the fault as timely as possible, a smaller range of 85 %CI is selected as the adaptive threshold among 85 %CI and 95 %CI.

4.2. Comparison of early warning performance with different methods

Early warning research based on field data is more valuable for avoiding thermal runaway accidents than laboratory-based methods. To verify the advanced nature of the proposed method, we compare our method with other algorithms applicable to field data in this section, including cluster-based (discrete Fréchet distance combined with local outlier factor, DFD + LOF) [29], deep

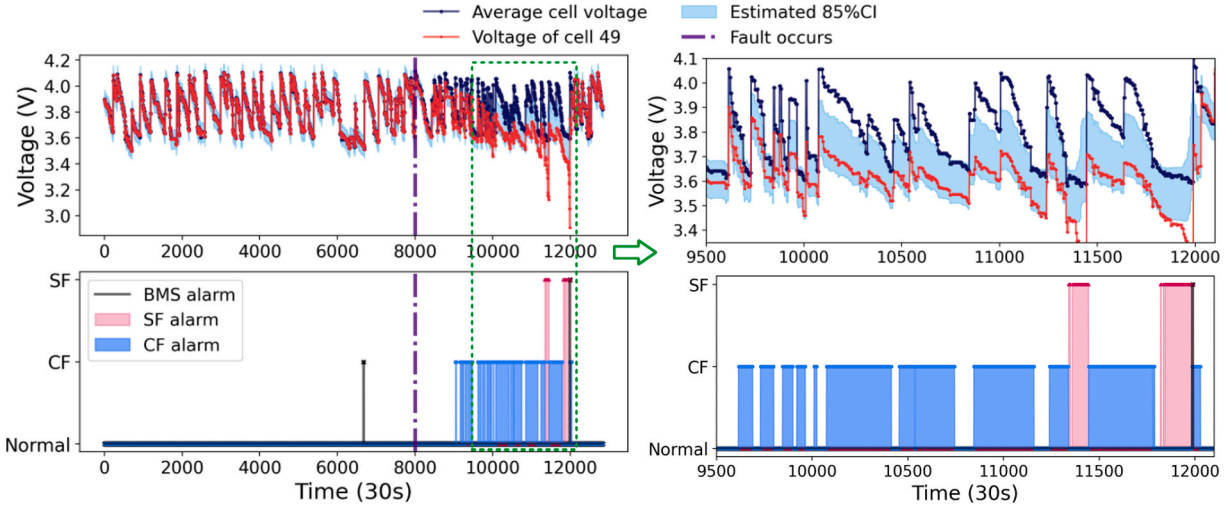


Fig. 5. Demonstration of the fault detection and location principle. The left plots depict real-time fault monitoring during vehicle operation. The right plot presents an enlarged view of the monitoring results when a fault is detected. Accurate fault location is achieved by simultaneously monitoring all the cells in the battery pack separately.

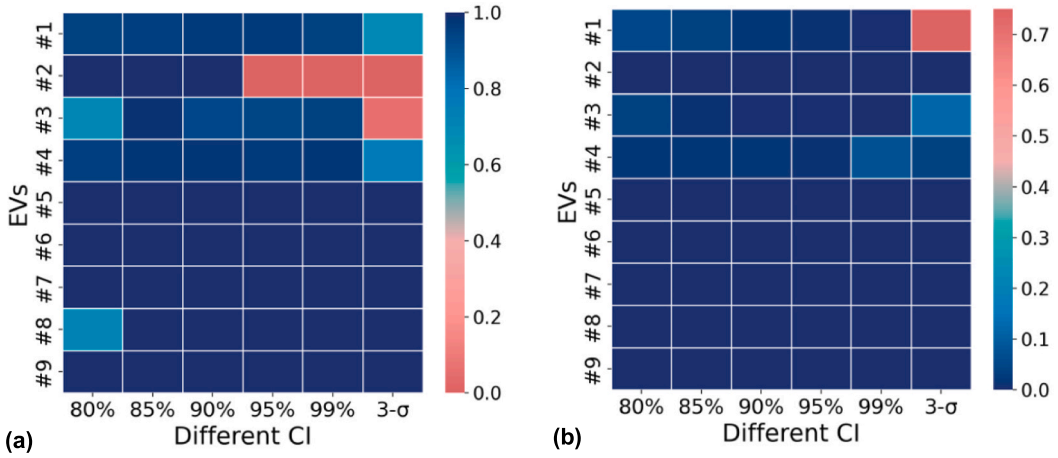


Fig. 6. Fault detection results with different CIs. (a) Precision. (b) FP rate.

learning-based (long short-term memory, LSTM) [30], and data-driven (state representation methodology, SRM) [31]. The parameter settings of the comparison methods all follow the corresponding reference. Moreover, the 3-rule-based method is not used for comparison due to its high FP rate (demonstrated in Section 4.1). In addition, some methods that are suitable for diagnosis based on field data but have special needs for data, such as methods that rely on charging data [16,32,33] or require a data sampling frequency no greater than 10 s [17,34], are not used for comparison in this study.

The early warning performances are summarized in Table 2. The proposed method realizes the earliest early warning for both developmental and sudden faults. It shows that it has superior ability of early fault detection compared with the comparison method. The contrasting methods also demonstrate their early-warning capabilities. However, in addition to the late warning time, there is also the problem of unstable warning performance. The variability in fault behaviors across different EVs contributes to this instability. These existing methods lack an efficient adaptive threshold akin to the proposed approach, which fails to dynamically adjust according to the operational conditions during the detection process. For developmental faults, the proposed method achieves 22.3–240.1 h early warning compared with the BMS. This is helpful for the timely maintenance of EVs and the elimination of hidden dangers. For sudden faults, the proposed method can provide 2–28 s early warnings, providing more time for fault tolerance control and personnel escape.

4.3. Safety risk assessment with a dynamic threshold

Battery safety is indeed a dynamic aspect that evolves throughout its service cycle. Visualizing the safety risk helps to quantify fault severity, track fault evolution, and provide guidance for EV maintenance. To accomplish this, we construct a safety risk assessment method that incorporates dynamic thresholds. This method enables a comprehensive assessment of the safety risk of batteries, considering three key aspects: voltage anomalies, cumulative risk, and temperature anomalies. In terms of voltage-based evaluation, we establish a fault-free threshold for the safe-risk function at 80 % CI. This threshold serves as a reference point to determine the normal operating range of a battery's voltage. Moreover, for identifying SFs, users have the flexibility to select a threshold such as 90 % CI, 95 % CI, or even 99 % CI, depending on the evaluation criteria. The dynamic threshold selection is based on the analysis of the fault detection performance with different CIs. For the temperature-based evaluation, the fault-free and SF thresholds of the safe-risk function are empirically set to 55 °C and 90 °C, respectively [26]. In this work, the weights of different aspects of safety impact are set equal for calculating the *Score*. When the *Score* of all cells is equal to 1, driving is completely safe. However, if the *Score* of one or more cells falls between 0.8 and 1, it signals that there are safety risks that affect the battery performance, and the EV should be repaired promptly. When the *Score* is less than 0.8, the battery poses a severe safety hazard, and it is crucial to immediately cease the operation of the EV to mitigate any potential dangers. A *Score* closer to 0 indicates a higher probability of thermal runaway of the battery.

Table 2
Early warning performance of different methods. Negative values indicate that the warning time lags behind the BMS alarm. FP indicates that a false alarm has occurred.

Method	Lead time of fault detection compared to BMS alarm								
	EV#1	EV#2	EV#3	EV#4	EV#5	EV#6	EV#7	EV#8	EV#9
DFD + LOF	14.4 h	5.0 h	199.1 h	6.5 h	4 s	3 s	6 s	4 s	2 s
LSTM	60.3 h	−16.7 h	139.8 h	15.1 h	3 s	5 s	20 s	3 s	2 s
SRM	13.3 h	8.4 h	151.6 h	21.9 h	2 s	5 s	0	2 s	1 s
Proposed	67.1 h	142.2 h	240.1 h	22.3 h	3 s	5 s	28 s	3 s	2 s

To clearly demonstrate the proposed safety risk assessment method, we present the evaluation results for two faulty cells (cell 2 and cell 49) of EV#1 as an illustrative example. Moreover, the SF threshold of the safety risk assessment function is initially set to 99 % CI. The evaluation process encompasses multiple perspectives and employs different evaluation criteria to assess the safety risk, as demonstrated in Fig. 7. The assessment result based on voltage anomalies (Fig. 7a) is consistent with the SF detection results. In Fig. 7b, the battery pack is evaluated for safety using the idea of risk accumulation (see Supplementary Fig. S4 for explanation). This method focuses on visualizing the evolution of battery faults and filters out sharp voltage variations. Notably, when the fault disappears, the *Score* of the faulty cells does not increase to 1 but remains unchanged. This is a reminder to the user or regulator that faulty cells need to be checked further because they are not as secure as are the normal cells. Furthermore, combined with the assessment results based on temperature anomalies (Fig. 7c), the final safety assessment results for EV#1 are shown in Fig. 7d. These results demonstrate that the proposed method provides reliable safety assessments without FPs. The *Score* of normal cells is affected only by temperature, and all remain at a safe level according to the voltage-based evaluation. Additionally, the variation trends of the *Score* and actual voltage of the fault cells are similar, illustrating that an effective fault evolution analysis is achieved.

By choosing different SF thresholds, our method can adapt different levels of safety assessment based on different criteria. Fig. 7e and f, which show two enlarged views, specifically demonstrate the performance using different CIs as SF thresholds for the safety assessment of CFs in cell 2 and SFs in cell 49, respectively. It can be observed that when a battery fault does not escalate into an SF, the selection of different SF thresholds has a slight impact on the safety evaluation performance. However, when the fault further evolves into an SF, the faulty cell's *Score* in the 90 % CI assessment as the SF threshold decreases to 0.8 more rapidly, and its *Score* remains at a low level after the SF occurs, indicating that the selection of a CI with a lower confidence level will result in more stringent safety assessment results. Moreover, more stringent safety assessments provide more conservative advice for EV maintenance to avoid thermal runaway accidents. To illustrate the generality, the verification results for other EVs are shown in Supplementary Fig. S5. In summary, our safety risk assessment method not only effectively tracks the evolution of battery faults, but also provides valuable guidance for EV maintenance.

4.4. Generalization validation based on laboratory data

To further verify the generality of the proposed method, we conducted ISC experiments of two series-connected battery modules. The ISC of batteries is one of the main causes of thermal runaway in EVs [35]. Each battery module consists of six Panasonic NCR18650BD cells, and the specifications of the cell can be seen from [36]. One of the cells simulates an ISC fault by connecting a resistor. To simulate the early stage of ISC, resistors of 500 Ω and 1000 Ω are chosen for the two battery modules, respectively. The experimental platform is DIGATRON battery test system. The operating conditions used are 1C constant current charge and dynamic stress test (DST) discharge. After reaching full charge, the battery module starts discharging after an hour of rest. A total of 16 cycles were performed for each battery module (8 normal cycles + 8 ISC cycles), all at an ambient temperature of 25 ℃. Moreover, the sampling interval is set to 10 s to simulate the field scenarios. The experimental schematic is presented in Supplementary Fig. S6.

In the validation, data from two random cycles in 0–5 cycles are used as training sets, and data of 6–16 cycles are used for testing.

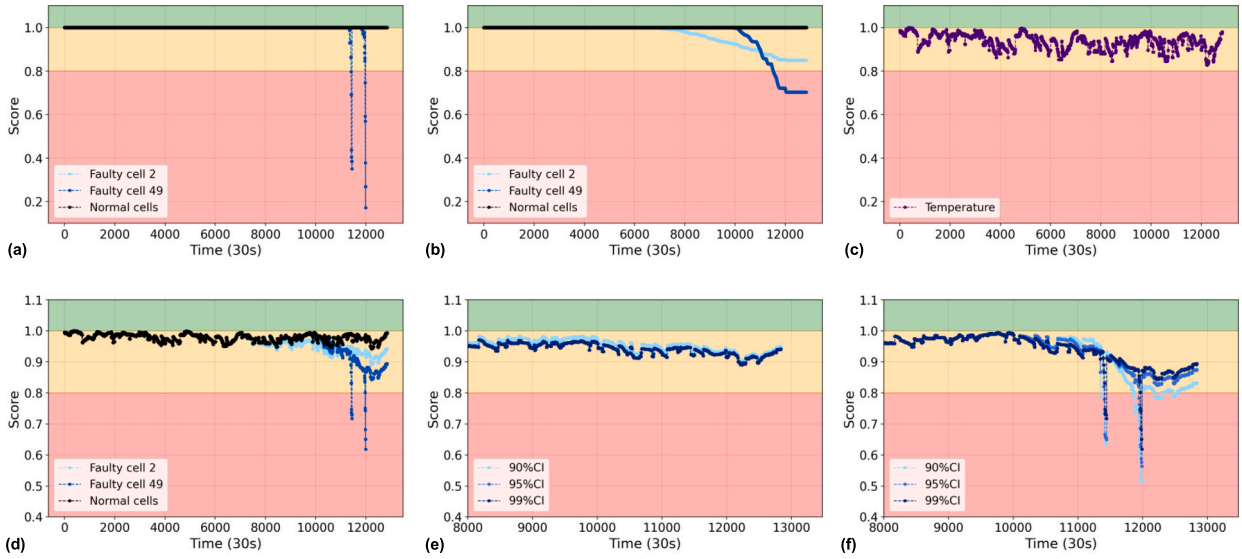


Fig. 7. Safety assessment results of EV#1 with different perspectives and criteria. The green, yellow, and red areas indicate the safety status of the battery at safe, warning, and dangerous levels, respectively. (a), (b), and (c) are the evaluation results based on voltage anomalies, the cumulative risk, and temperature anomalies, respectively. (d) Comprehensive results of the safety assessment. (e) Assessment result of cell 2 based on different criteria. (f) Assessment result of cell 49 based on different criteria. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The input features and the training process are set up as in the field data validation. Moreover, since the temperature of cells was not collected in the experiment, only the voltage anomaly is analyzed in the safety risk assessment. The validation results are shown in Fig. 8. It can be seen that the fault is detected at about 2 and 3 cycles after the fault injection for cells with 500 Ω and 1000 Ω resistors, respectively. In the safety risk assessment, since the cell with 500 Ω experiences a larger short-circuit current, its *Score* eventually lower than that of the cell with 1000 Ω and decreases faster. Moreover, the *Score* of normal cells are always maintained at 1, indicating that there is no FP. Therefore, the proposed method is effective for the detection and assessment of ISC faults in laboratory scenarios.

To prove the advancement of the proposed method, the fault detection evaluations of the three comparison methods are conducted (as used in Section 4.2, including DFD + LOF, LSTM and SRM). The training of the LSTM model is based on the first three fault-free cycles, consistent with that of the proposed method. The verification results of these comparison methods are illustrated in Fig. 9. The analysis reveals that the DFD + LOF method exhibits weak sensitivity towards early faults. In the verification of detecting injected fault with 1000 Ω resistor, it is unable to distinguish between faulty and non-faulty battery cells. Moreover, due to the dependence on a fixed threshold, the ability of LSTM and SRM methods to balance early detection and FPs is limited. They all showed FPs in detecting injected faults with 500 Ω resistor. In contrast, the proposed method shows notable advantages. It has a shorter fault detection time post-injection, devoid of both FPs and false negatives, primarily attributed to its probabilistic adaptive threshold mechanism. This enhancement ensures its superior performance when compared with the traditional methods.

4.5. Generalization validation based on open-source data

To further verify the generalization of the proposed method, the field data of 28 faulty battery systems (BSs) provided by the University of Darmstadt are used for validation in this section (they are denoted as BS #1-#28 in this paper) [37]. These BSs are equipped with lithium iron phosphate batteries with a rated capacity of about 160 Ah. Each BS consists of eight cells in series, with individual current and voltage sensors allocated per cell. Additionally, four temperature sensors are positioned between adjacent cells, with each temperature sensor serving two cells. The specific use of these portable BSs is unknown, and they are all returned to the manufacturer because they show some unsatisfactory behavior. The voltage and temperature variation curves for each BS are shown in Supplementary Fig. S7, showcasing instances of irregular undervoltage or overvoltage occurrences across these BSs. The causes of these anomalies are varied and may include short circuits, mistreatment, sensor malfunctions, among other factors. Notably, due to active equalization mechanisms or manual interventions, some anomalies transiently revert to normal operating levels.

Compared with the field data of EVs used in this work, these BS data exhibit two primary distinctions. On the one hand, the anomalies within the BSs may not solely stem from battery faults, and crucially, the BMS alarm information is absent in the dataset, precluding the evaluation of early fault warning. On the other hand, the dataset lacks explicit fault labels assigned to each time point, and the active repair measures employed disrupt the typical fault deterioration trend over time. Therefore, during the verification process, the strategy of using data with high voltage consistency for model training and evaluating the full-cycle safety risks is adopted. Because the cell voltage maintains a high degree of consistency can usually be regarded as the battery is fault-free [38]. To mitigate the

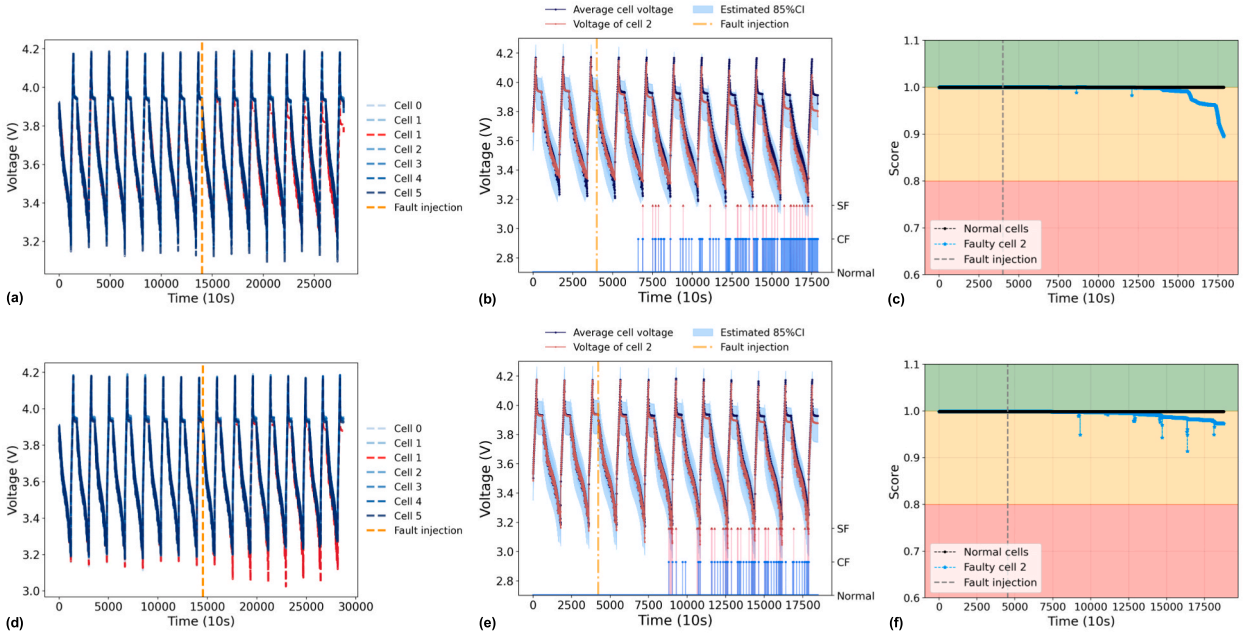


Fig. 8. The validation results of laboratory data. (a) Voltage variation with 500 Ω resistor. (b) Fault detection result with 500 Ω resistor. (c) Safety assessment result with 500 Ω resistor. (c) Voltage variation with 1000 Ω resistor. (d) Fault detection result with 1000 Ω resistor. (e) Safety assessment result with 1000 Ω resistor.

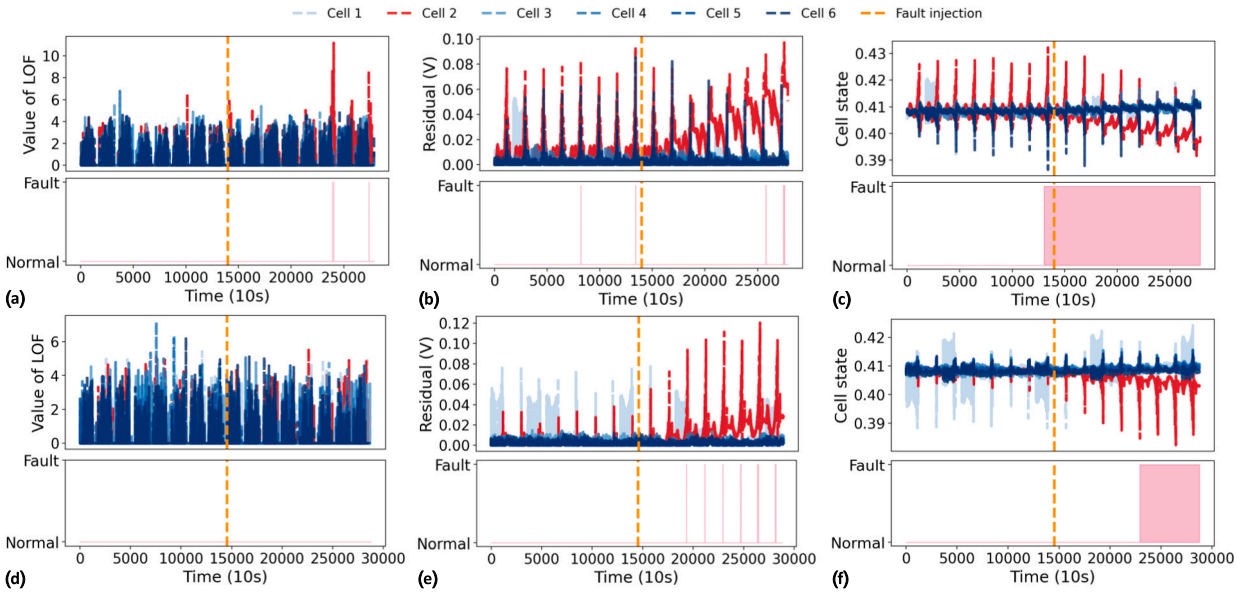


Fig. 9. Fault detection results of the comparison methods. (a), (b) and (c) are the fault detection results of with 500 Ω resistor based on DFD + LOF, LSTM, and SRM methods, respectively. (a), (b) and (c) are the fault detection results of with 1000 Ω resistor based on DFD + LOF, LSTM, and SRM methods, respectively.

influence of proactive maintenance on safety risk assessment, an assessment strategy of voltage difference rate based on dynamic boundaries is adopted in safety risk assessment, rather than relying on accumulated voltage differences over time.

Fig. 10 shows the density analysis results of the cell voltage to roughly visualize the voltage consistency. It shows that the standard deviation of most cell voltages is within 0.01. Therefore, data with a standard deviation of cell voltage less than 0.01 are selected for model training in this work. To ensure the fairness of the verification, the characteristics of the input model are the same as those of the verification EV. Moreover, the amount of training data is randomly selected with a period of 5 days (the sampling interval of the BS data is 5 s, resulting in 86,400 selected training data points). The detection examples of CF and SF based on the proposed method are shown in Fig. 11. It can be seen that when SF is detected, the cells all exhibit obvious overvoltage or undervoltage behavior. No misjudged SF occurred when the voltage is within the normal operating range. Furthermore, the safety risk of all BSs is evaluated using the proposed method, with results depicted in Supplementary Fig. S8. The result reveals that the computed *Score* effectively mirrors the aberrant behaviors exhibited by cells, indicating a robust capability to discern anomalies accurately. Furthermore, through the integration of voltage inconsistencies and temperature considerations, the proposed method enables a comprehensive evaluation of the safety risks associated with the BSs. Overall, despite the divergent applications of these BSs compared to EVs, the proposed methodology demonstrates efficacy in detecting and assessing faults within portable BSs.

5. Conclusion

Timely and accurate detection and tracking of battery faults play crucial roles in guiding the maintenance of EVs and minimizing

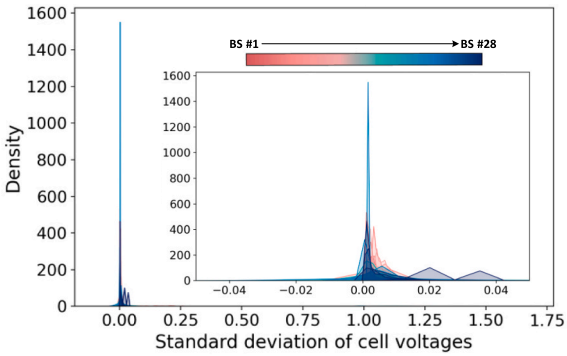


Fig. 10. The density analysis results of the standard deviation of cell voltages.

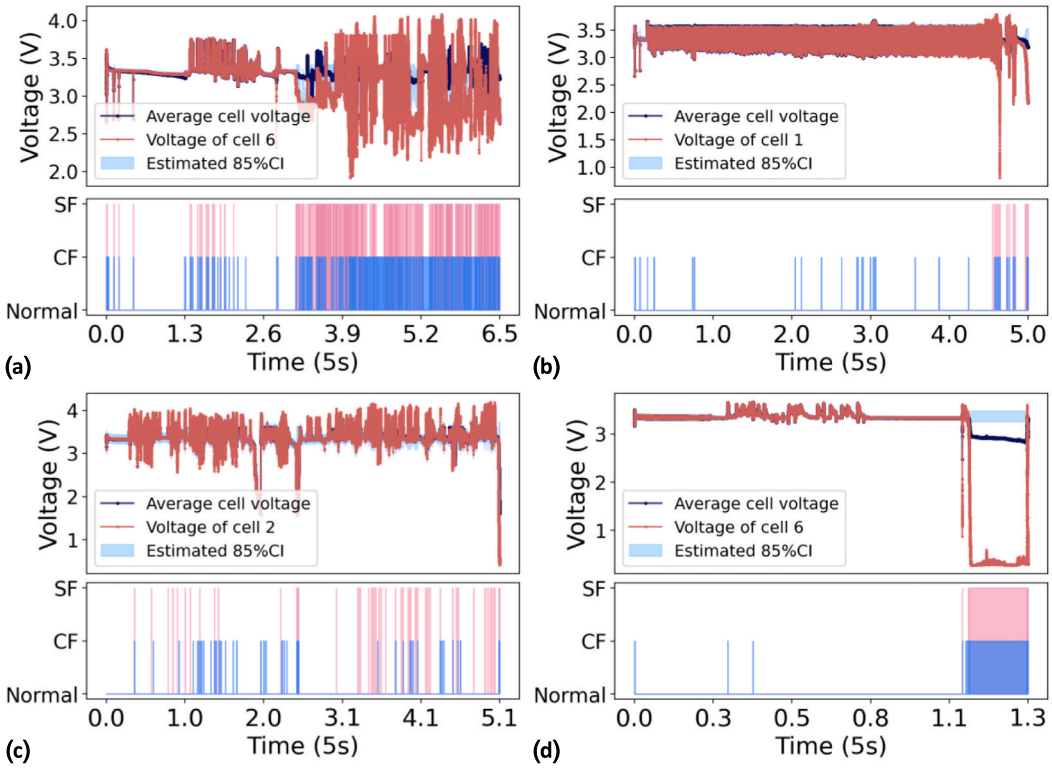


Fig. 11. Example of fault detection results. (a) Cell 6 of BS #1. (b) Cell 1 of BS #9. (c) Cell 2 of BS #17. (d) Cell 6 of BS #27.

the occurrence of thermal runaway accidents. To meet this goal, we propose a probability-based fault detection and safety evaluation method in this study. This method leverages real-time GPR-based CI to adapt to fault thresholds, enhancing the flexibility and adaptability of the fault detection process. The effectiveness of this method is verified using a dataset of 1046 cells from nine faulty EVs, two battery modules and twenty-eight battery systems. The proposed method achieves accurate fault detection with an average precision and FP rate of 0.99 and 0.07, respectively. For field data, it provides early warning for developmental and sudden faults compared to BMS alarm 22.3–240.1 h and 2–28 s, respectively. One of the advantages of this method is its suitability for different field data. This method requires only a short duration of historical data, specifically five days, to construct an effective fault detection and location model. Moreover, we introduce a real-time safety assessment method to track the evolution trend of battery faults. This approach enables the visualization of the safety risk of each cell, enriches the fault detection performance, and can provide valuable insights for battery maintenance and insurance. Furthermore, the method demonstrates the flexibility of achieving different levels of safety evaluation by selecting different confidence thresholds. Additionally, the efficacy of the proposed approach extends beyond electric vehicle (EV) applications, proving valuable in laboratory settings and for fault detection in portable battery systems as well.

In the future, we will integrate more battery physical information into the safety evaluation framework, such as the capacity, internal resistance and electrochemical parameters of the battery, to enhance the guidance provided for battery maintenance and battery manufacturing.

CRediT authorship contribution statement

Jinwen Li: Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Yunhong Che:** Writing – original draft, Software, Methodology, Investigation. **Kai Zhang:** Methodology, Data curation. **Jia Guo:** Validation, Formal analysis, Data curation. **Hongao Liu:** Visualization, Investigation. **Yi Zhuang:** Visualization, Software, Investigation. **Congzhi Liu:** Supervision, Conceptualization. **Xiaosong Hu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Remus Teodorescu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China under grant number U23A20327, the National Key Research and Development Program of China under grant number 2022YFB3305403, the project of basic research funds for central universities under grant number 2023CDJQCZX-001 and 2022CDJDX-006, and the National Natural Science Foundation of China under grant number 72361137006.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ymssp.2025.112951.

Data availability

Data will be made available on request.

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